
Artificial Intelligence

Tut/Lab week 5

COSC2129/3118

Outline

- Highlights:
 - Assignment 1
 - FOL
 - Inference rules
 - Forward and backward inference
- Q&A

Assignment 1

- Part 1:
 - Most of the solutions are well done (understand the algorithms, implementation & scores)
 - Some of the questions are not properly implemented, or with low or 0 score
 - You still get some points if you have the code implemented but with low score.
 - Use AI?
- Part 2: Question?

Materials

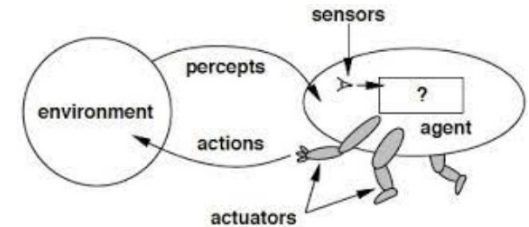
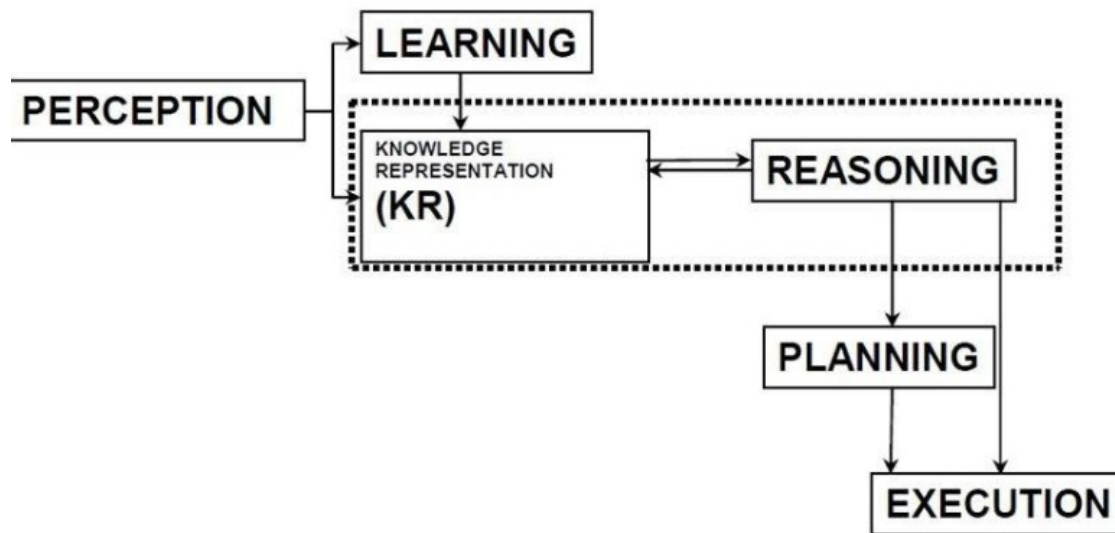
- Book: Artificial Intelligence: A Modern Approach
- Chapter 7 [7.8, 7.9]

Knowledge & Reasoning

- Knowledge representation: How do humans represent and store knowledge?
- Reasoning: How do humans do reasoning?
- How to transfer human knowledge to an agent?

Knowledge & Reasoning

The AI Cycle



Knowledge & Reasoning

- To be intelligent, an agent needs a **knowledge base** to store knowledge and an **inference engine** to do the reasoning.
- 2 types of logic:
 - **Propositional Logic (PL)**
 - **First-order Logic (FOL)**
- What are the desirable properties of a knowledge representation (KR) scheme?

Knowledge representation in FOL

- **Constant symbols** represent individuals in the world
- **Predicates symbols** map individuals to truth values
 - `greater(3>2)`
 - `color(Grass, Green)`
- **Function symbols**, which map individuals to individuals
 - `Sqrt(4)`
 - `father-of(Daisy) = John`
- **Variable symbols**
 - E.g., $x, y, \dots$
- **Connectives**
 - Same as in PL: not ($\neg$), and ($\wedge$), or ($\vee$), implies ($\rightarrow$), if and only if (biconditional $\leftrightarrow$)
- **Quantifiers**
 - Universal $\forall x$ or **(Ax)**
 - Existential $\exists x$ or **(Ex)**
- **Equality:** =

Examples

- How can you write sentences, facts,... in FOL?
- First-order logic (FOL) models the world in terms of
 - **Objects**, which are things with individual identities
 - **Properties** of objects that distinguish them from other objects
 - **Relations** that hold among sets of objects
 - **Functions**, which are a subset of relations where there is only one “value” for any given “input”
- Example: See Tutorial 05 - Exercies

- Example: “all zebras are striped” in FOL
 - $\forall x \text{ zebra}(x) \Rightarrow \text{striped}(x)$
 - $\forall x \text{ zebra}(x) \wedge \text{striped}(x)$ (?)
- Note the meanings:
 - “for all x in the universe of discourse, if x is a zebra, then x is striped”
 - “everything is a zebra and everything is striped, meaning everything in the universe of discourse is a striped zebra”!

- How about

$$\exists x \text{ zebra}(x) \Rightarrow \text{striped}(x)$$

- there exists an x, where if x is a zebra, then x is striped

$$\exists x \text{ zebra}(x) \wedge \text{striped}(x)$$

- there exists an x, where x is a zebra and x is striped.

- Every researcher published at least one research paper.

$$\forall x \text{ researcher}(x) \Rightarrow \exists y (\text{paper}(y) \wedge \text{pub}(x,y))$$

Inference in FOL

- Propositionalization: **FOL $\rightarrow$ PL**

Inference in FOL is an extension of inference in PL.

We can reduce FOL inference to propositional inference via **propositionalization**.

(Need to deal with *substitutions* and *quantifiers*)

The basic idea

- A variable of sentences with existential quantifier can be substituted by one instantiation.
- A variable of sentences with universal quantifier can be substituted by the set of *all possible* instantiations.

Generalized Modus Ponens

Modus Ponens can be extended (lifted) for inference in FOL – **Generalized Modus Ponens**.

$$\frac{p'_1, p'_2, \dots, p'_n, (p_1 \wedge p_2 \wedge \dots \wedge p_n) \Rightarrow q}{SUBST(\theta, q)}$$

where θ is a substitution such that for all i , $SUBST(\theta, p'_i) = SUBST(\theta, p_i)$

Substitution is written as: $\{ \langle \text{variable} \rangle / \langle \text{instantiation} \rangle \}$.

E.g., $\forall x \text{ Man}(x) \Rightarrow \text{Mortal}(x)$, Substitutions: $\{x/\text{Socrates}\}$, $\{x/\text{John}\}$, $\{x/\text{Richard}\}$, ...

$\exists x \text{ Man}(x) \wedge \text{Mortal}(x)$, Substitution: $\{x/M1\}$

- Modus Ponens + to find a substitution for the variables in sentences.

Skolemization & Unification

- **Unification** is a process of *making two different logical atomic expressions identical* by finding a substitution.
 - In other words: process of solving equations between symbolic expressions.
- **Skolemization** is a transformation on first-order logic formulae, which *removes all existential quantifiers* from a formula.
 - This technique is vital in proof theory and automated reasoning, especially for refutation based, like resolution.

Skolemization

Conversion of sentences FOL to CNF requires skolemization.

Skolemization: remove existential quantifiers by introducing new function symbols.

How: For each existentially quantified variable introduce a n -place function where n is the number of previously appearing universal quantifiers.

Special case: introducing constants (trivial functions: no previous universal quantifier).

Skolemization

Every philosopher writes at least one book.

$$\forall x[Philo(x) \rightarrow \exists y[Book(y) \wedge Write(x, y)]]$$

Eliminate Implication:

$$\forall x[\neg Philo(x) \vee \exists y[Book(y) \wedge Write(x, y)]]$$

Skolemize: substitute y by $g(x)$

$$\forall x[\neg Philo(x) \vee [Book(g(x)) \wedge Write(x, g(x))]]$$

FOL $\rightarrow$ PL

- How about Universal quantifier $\forall x$?

Most General Unifier (MGU)

- Two atomic sentences: pair up the formulas and align them with each other.

We can compute the MGU using the disagreement set $D_k = \{e_1, e_2\}$: the pair of expressions where two clauses first disagree.

REPEAT UNTIL no more disagreement $\rightarrow$ found MGU.

IF either e_1 or e_2 is a variable V and the other is some term (or a variable) t , then choose $V = t$ as substitution.

Then substitute to obtain S_{k+1} and find disagreement set D_{k+1} .

ELSE unification is not possible.

Most General Unifier (MGU)

- Is the most common generalization of two atomic sentences.
 - To calculate MGU: pair up the formulas and align them with each other.
 - Example:
 - $P(x, f(y)) \ \& \ P(a, f(g(z)))$
 - $S_0 =$
 - $D_1 =$
 - $\text{Theta}_1 =$
 - ...
- $\Rightarrow \text{theta} = [x/a, y/g(z)] \quad (\text{substitution})$

MGU

MGU: $[t/u]$

- $f(t_1, \dots, t_n) / f(u_1, \dots, u_n)$, or $[t_1/u_1, \dots, t_n/u_n]$
- $f(t_1, \dots, t_n) / f(u_1, \dots, u_m)$, if $f \neq f$ or $n \neq m$ then fails.
- if x/x : remove and continues
- $x/f(t_1, \dots, t_n)$
- $f(t_1, \dots, t_n) / x$ then replace x with $f(t_1, \dots, t_n)$
- If x is a part of $t_1, \dots, t_n$ then fails.

Most General Unifier (MGU)

- Example:
 - $Q(a, g(x,a), f(y)) = Q(a, g(f(b),a), x)$
 $\Rightarrow \theta = ?$
 - $a \sim a$
 - $x \sim f(b)$
 - $y ?$
 - $\theta = [x/f(b), y/b]$
- $P(x, f(y)) \& P(a, f(g(z)))$
 - $\theta = [x/a, g(b)/y, b/z]$

KR & Inference in FOL

The approach is:

1. decide predicates and functions
2. construct a knowledge base
3. search for a proof

By forward-chaining:

-- starts with known facts and infer new ones until reaching the goal

Or by backward-chaining:

-- starts with the goal, and work backwards until hitting known facts

Hand-on Practice

- [AI22 Tut5](#)

Question?